Graphs 6 — Cubic graphs · Reciprocal graphs

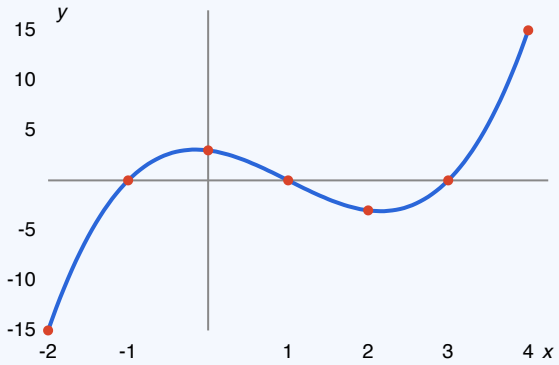
Calculator allowed; graph paper needed. Marks in brackets. ★ = harder question. Plot the points and join them with a *smooth* curve — never with a ruler.

1 **a** Copy and complete the table of values for $y = x^3 - 3x^2 - x + 3$, then draw the graph for $-2 \le x \le 4$. **b** Write down the values of x where the graph cuts the x-axis. [6]

x	-2	-1	0	1	2	3	4
y	-15		3		-3		15

a Substitute carefully, cubing negatives: $(-1)^3 = -1$.

x	-2	-1	0	1	2	3	4
y	-15	0	3	0	-3	0	15



A positive cubic: down on the left, up on the right, with a "hill" and a "valley". **b** The graph cuts the x-axis at $x = -1$, $x = 1$ and $x = 3$ (where $y = 0$).

2 The depth of water D metres in a tidal pool, t hours after 6 a.m., is given by $D = t^3 - 6t^2 + 9t + 2$ for $0 \leq t \leq 4$. [9]

a Copy and complete the table and draw the graph of D against t.

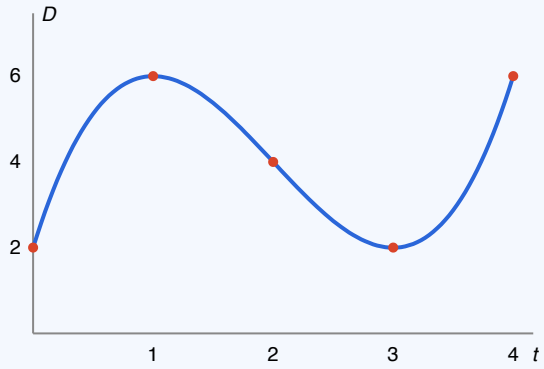
b Use your graph to estimate the greatest depth and the time when it occurs.

c Between which times is the depth less than 3 m?

t (hours)	0	1	2	3	4
D (m)	2		4		6

a

t (hours)	0	1	2	3	4
D (m)	2	6	4	2	6



b The curve peaks at $t = 1$: greatest depth 6 m, at 7 a.m. (It dips to 2 m at $t = 3$, 9 a.m., then rises again.)

c Draw the line $D = 3$ and read where the curve is below it: from about $t = 2.4$ to $t = 3.5$, i.e. roughly 8:20 a.m. to 9:30 a.m. (Any sensible reading from an accurate graph is accepted — e.g. $2.3 \leq t \leq 3.6$.)

3 The cost £y per item of producing x hundred items is modelled by $y = \frac{x}{2} + \frac{8}{x}$ for $1 \le x \le 8$. [10]

a Copy and complete the table (2 d.p.).

b Draw the graph for $1 \le x \le 8$, with y from 0 to 9.

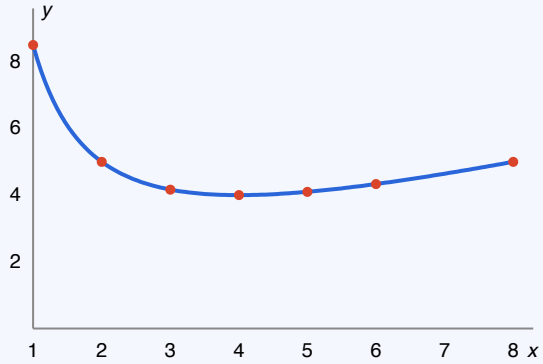
c Use your graph to estimate the cost per item when 350 items are made.

d How many items should be made for the lowest cost per item, and what is that cost?

x	1	2	3	4	5	6	8
$\frac{x}{2}$	0.5		1.5		2.5		4
$\frac{8}{x}$	8		2.67		1.6		1
y	8.50		4.17		4.10		5.00

a

x	1	2	3	4	5	6	8
$\frac{x}{2}$	0.5	1	1.5	2	2.5	3	4
$\frac{8}{x}$	8.00	4.00	2.67	2.00	1.60	1.33	1.00
y	8.50	5.00	4.17	4.00	4.10	4.33	5.00



b The curve falls steeply, flattens near $x = 4$, then rises slowly — the $\frac{8}{x}$ term dominates for small x, the $\frac{x}{2}$ term for large x.

c 350 items is $x = 3.5$: reading from the graph $y \approx 4.0$ (exact: $1.75 + 2.29 = 4.04$), about £4.04 per item.

d The minimum is at $x = 4$, i.e. 400 items, costing $\frac{4}{2} + \frac{8}{4} = £4$ per item.

Total: 25 marks

Check: build a table row by row (x^3, $-3x^2$, …) and add · cubes of negatives are negative · smooth curve through every point, no straight segments · a reciprocal graph never touches $x = 0$ · read a graph by drawing a horizontal or vertical line and marking the crossing.